Practical 15

Write JavaScript code for Validating the various Form Elements

index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Practical 15 - Form Validation</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f2f2f2;
            padding: 20px;
        }

        section {
            background-color: white;
            padding: 25px;
            border: 2px solid #333;
            border-radius: 10px;
            max-width: 600px;
            margin: auto;
        }

        .box {
            background-color: #eeeeee;
            padding: 20px;
            border-radius: 8px;
        }

        label {
            display: block;
            margin-top: 12px;
        }

        input, select, button {
            padding: 10px;
            margin-top: 5px;
        }
```

```
        input[type="radio"],
        input[type="checkbox"] {
            margin-right: 5px;
        }

        #result {
            color: blue;
            font-weight: bold;
        }
    </style>
</head>

<body>
    <section>
        <h1>Practical 15</h1>
        <h2>Form Validation</h2>

        <div class="box">
            <form id="myForm">
                <label>Enter Name:</label>
                <input type="text" id="name">

                <label>Enter Email:</label>
                <input type="text" id="email">

                <label>Enter Password:</label>
                <input type="password" id="password">

                <label>Enter Phone Number:</label>
                <input type="text" id="phone">

                <label>Select Gender:</label>
                <input type="radio" name="gender" value="Male"> Male
                <input type="radio" name="gender" value="Female"> Female

                <label>Select Course:</label>
                <select id="course">
                    <option value="">Select Course</option>
                    <option value="BSc AI">BSc CS</option>
                    <option value="BCA">BSc AI</option>
                    <option value="BSc IT">BSc IT</option>
                </select>
```

```
            <br><br>

            <input type="checkbox" id="terms">
            I accept the terms and conditions.

            <br><br>

            <button type="submit">Submit</button>
        </form>

        <p id="result"></p>
    </div>
</section>

<script>
    document.getElementById("myForm").addEventListener("submit",
function(event) {
        event.preventDefault();

        let name = document.getElementById("name").value.trim();
        let email = document.getElementById("email").value.trim();
        let password = document.getElementById("password").value;
        let phone = document.getElementById("phone").value.trim();
        let course = document.getElementById("course").value;
        let gender =
document.querySelector('input[name="gender"]:checked');
        let terms = document.getElementById("terms").checked;

        if (name === "") {
            document.getElementById("result").innerHTML =
                "Please enter your name.";
        }
        else if (!/^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(email)) {
            document.getElementById("result").innerHTML =
                "Please enter a valid email.";
        }
        else if (password.length < 6) {
            document.getElementById("result").innerHTML =
                "Password must contain at least 6 characters.";
        }
        else if (!/^\d{10}$/.test(phone)) {
```

```
                document.getElementById("result").innerHTML =
                    "Please enter a valid 10-digit phone number.";
            }
            else if (!gender) {
                document.getElementById("result").innerHTML =
                    "Please select your gender.";
            }
            else if (course === "") {
                document.getElementById("result").innerHTML =
                    "Please select a course.";
            }
            else if (!terms) {
                document.getElementById("result").innerHTML =
                    "Please accept the terms and conditions.";
            }
            else {
                document.getElementById("result").innerHTML =
                    "Form submitted successfully!";
            }
        });
    </script>
</body>
</html>
```

Practical 16

Write Javascript code for

a. Demonstrating different JavaScript Objects such as Window, Navigator, History, Location, Document, Storing and Retrieving Cookies

index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Practical 16 - JavaScript Objects and Cookies</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f2f2f2;
            padding: 20px;
        }

        section {
            background-color: white;
            padding: 25px;
            border: 2px solid #333;
            border-radius: 10px;
            max-width: 800px;
            margin: auto;
        }

        .box {
            background-color: #eeeeee;
            padding: 15px;
            margin: 15px 0;
            border-radius: 8px;
        }

        button, input {
            padding: 10px;
            margin: 5px;
        }

        button {
            cursor: pointer;
        }
```

```
        p {
            color: blue;
            font-weight: bold;
            overflow-wrap: anywhere;
        }
    </style>
</head>

<body>
<section>
    <h1>Practical 16</h1>
    <h2>JavaScript Objects and Cookies</h2>

    <div class="box">
        <h3>1. Window Object</h3>
        <button onclick="windowExample()">Show Details</button>
        <p id="windowResult"></p>
    </div>

    <div class="box">
        <h3>2. Navigator Object</h3>
        <button onclick="navigatorExample()">Show Details</button>
        <p id="navigatorResult"></p>
    </div>

    <div class="box">
        <h3>3. History Object</h3>
        <button onclick="historyExample()">Show Details</button>
        <p id="historyResult"></p>
    </div>

    <div class="box">
        <h3>4. Location Object</h3>
        <button onclick="locationExample()">Show Details</button>
        <p id="locationResult"></p>
    </div>

    <div class="box">
        <h3>5. Document Object</h3>
        <button onclick="documentExample()">Show Details</button>
        <button onclick="changeTitle()">Change Title</button>
```

```
        <p id="documentResult"></p>
    </div>

    <div class="box">
        <h3>6. Storing and Retrieving Cookies</h3>

        <input type="text" id="username"
               placeholder="Enter your name">

        <br>

        <button onclick="storeCookie()">Store Cookie</button>
        <button onclick="retrieveCookie()">Retrieve Cookie</button>

        <p id="cookieResult"></p>
    </div>
</section>

<script>
    // 1. Window Object
    function windowExample() {
        document.getElementById("windowResult").innerText =
            "Width = " + window.innerWidth +
            " pixels, Height = " + window.innerHeight + " pixels";
    }

    // 2. Navigator Object
    function navigatorExample() {
        document.getElementById("navigatorResult").innerText =
            "Language = " + navigator.language +
            ", Online = " + navigator.onLine;
    }

    // 3. History Object
    function historyExample() {
        document.getElementById("historyResult").innerText =
            "History Entries = " + history.length +
            ". Use browser Back to visit the previous page.";
    }

    // 4. Location Object
    function locationExample() {
```

```
    document.getElementById("locationResult").innerText =
        "URL = " + location.href +
        ", Protocol = " + location.protocol +
        ", Hostname = " + location.hostname;
}

// 5. Document Object
function documentExample() {
    document.getElementById("documentResult").innerText =
        "Title = " + document.title +
        ", URL = " + document.URL;
}

function changeTitle() {
    document.title = "Title Changed";

    document.getElementById("documentResult").innerText =
        "Page title changed successfully.";
}

// 6. Store Cookie
function storeCookie() {
    let name = document.getElementById("username").value.trim();
    let result = document.getElementById("cookieResult");

    if (name === "") {
        result.innerText = "Please enter your name.";
        return;
    }

    try {
        document.cookie =
            "username=" + encodeURIComponent(name) +
            "; max-age=86400; path=/; SameSite=Lax";

        if (document.cookie.split("; ").some(function(cookie) {
            return cookie.startsWith("username=");
        })) {
            result.innerText = "Cookie stored successfully!";
        } else {
            result.innerText =
                "Cookies are blocked.";
```

```
            }
        } catch (error) {
            result.innerText = "Unable to store the cookie.";
        }
    }

    // Retrieve Cookie
    function retrieveCookie() {
        let result = document.getElementById("cookieResult");
        let cookies = document.cookie.split("; ");

        for (let cookie of cookies) {
            let parts = cookie.split("=");
            let key = parts.shift();

            if (key === "username") {
                result.innerText =
                    "Stored Username = " +
                    decodeURIComponent(parts.join("="));
                return;
            }
        }

        result.innerText =
            "No cookie found. Store a cookie first.";
    }
</script>

</body>
</html>
```